

Assignment-2

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• Weather Forecasting:-

→ Weather forecasting is something we all rely on from knowing if we need an umbrella tomorrow to preparing for serious events like hurricanes or floods. Behind the scenes, it involves a lot of computing power and communication betⁿ different systems around the world.

• How Parallel Computing Helps:-

→ To predict the weather, scientists use complex computer models that simulate how the atmosphere behaves. These models involve tons of calculations far too many for a regular computer to handle quickly. That's where parallel computing comes in. Instead of using one processor to do all the work, parallel computing uses many processors at the same time to share the load. This helps meteorologists run their weather models faster and more accurately, which is super important when time is critical.

• How Networked Systems Helps:-

Weather data comes from all over the globe - satellites, airplanes, weather stations, and even ocean buoys. Networked systems are what connect all these sources, allowing the data to be sent in real time to weather centers. After the forecasts are made the same networks are used to send the information back out to news stations, weather apps, airports, and emergency services.

• Why It Matters:-

Using both parallel computing and networked system together allows us to get reliable and timely weather updates. This helps people plan their day, farmers protect their crops, airlines avoid dangerous conditions, and communities prepare for natural disasters.